Chapter: 4 Discrete Filter Structure

* $h[n] = 0$ for $n \geq M$ ← (Def^n : FIR)

Causal FIR:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k] \quad \{\text{original LTI system}\}$$

$$= \sum_{k=0}^{M-1} h[k]\, x[n-k]$$

$$= \sum_{k=0}^{M-1} b_k\, x[n-k]$$

(b_k → multiplier of b_k)

* Causal FIR Filter Realization

$$y[n] = \sum_{k=0}^{M-1} b_k\, x[n-k]$$

$$\Rightarrow \frac{Y(z)}{X(z)} = H(z) = \sum_{k=0}^{M-1} b_k\, z^{-k}$$

(2) Cascade - Form:

$$H(z) = \sum_{k=0}^{M-1} b_k\, z^{-k}$$

Modify : $H(z) = \prod_{k=1}^{k} H_k(z)$

where,

$H_k(z) = b_{k_0} + b_{k_1} z^{-1} + b_{k_2} z^{-2}$ &

k = integer part of $\frac{M+1}{2}$

(3) Lattice structure:

$H(z)$ = given

$H_m(z) = A_m(z)$, m = order of FIR filter

$A_m(z) = A_{m-1}(z) + K_m z^{-1} B_{m-1}(z)$

$B_m(z) = K_m A_{m-1}(z) + z^{-1} B_{m-1}(z)$

Solve: $A_{m-1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2}$

$B_m(z) = z^{-m} A_m(z^{-1})$

K_m = Lattice coefficient = $\alpha_m(m)$

↳ coefficient / multiplier of z^{-m} in $A_m(z)$.

if $K_1, K_2 \dots K_n$ given,

$$A_0(z) = B_0(z) = 1$$
$$A_m(z) = A_{m-1}(z) + K_m z^{-1} B_{m-1}(z)$$
$$B_m(z) = K_m A_{m-1}(z) + z^{-1} B_{m-1}(z)$$

IIR Filter Structure:

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

$$= H_1(z) \cdot H_2(z)$$

where, $H_1(z) = \sum_{k=0}^{M} b_k z^{-k}$

$$H_2(z) = \frac{1}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

Cascade Structure

$$H(z) = \prod_{k=1}^{P} H_k(z)$$

where,

$$H_k(z) = \frac{b_{k0} + b_{k1} z^{-1} + b_{k2} z^{-2}}{1 + a_{k1} z^{-1} + a_{k2} z^{-2}}$$

& P = integer part of $\left(\frac{N+1}{2}\right)$

(3) Lattice OR Lattice - Ladder structure:

$$H(z) = \frac{P(z)}{Q(z)}$$

Ladder:

Let $A_m(z) = Q(z) \Rightarrow$ Do as in FIR

$C_{m-1}(z) = C_m(z) - \underbrace{\delta_m}_{\text{ladder coefficient}} B_m(z)$, $\delta_m = C_m(m)$

where $C_m(z) = P(z)$.

Chapter : 5

FIR Filter Design

Common Window:

1. Rectangular : $w[n] = \begin{cases} 1, & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}$

2. Bartlett : $w[n] = \begin{cases} 2n/M, & 0 \le n \le M/2 \\ 2 - 2n/M, & M/2 < n \le M \\ 0, & \text{otherwise} \end{cases}$

3. Hanning : $w[n] = \begin{cases} 0.5 - 0.5\cos\left(\frac{2\pi n}{M}\right), & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}$

4. Hamming : $w[n] = \begin{cases} 0.54 - 0.46\cos\left(\frac{2\pi n}{M}\right), & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}$

5. Blackman : $w[n] = \begin{cases} 0.42 - 0.5\cos\left(\frac{2\pi n}{M}\right) + 0.08\cos\left(\frac{4\pi n}{M}\right), & 0 \le n \le M \\ 0, & \text{otherwise.} \end{cases}$

CamScanner

	Main lobe width	Peak Approx. Error or Stopband attenuation [dB]
1.	$4\pi/(M+1)$	-21
2.	$8\pi/M$	-25
3.	$8\pi/M$	-44
4.	$8\pi/M$	-53
5.	$12\pi/M$	-74

Linear-Phase Low Pass FIR Filter:

Desired freq. response: $H_{lp}(j\omega) = H_{lp}(e^{j\omega})$

$$= \begin{cases} e^{-j\omega M/2}, & |\omega| \le \omega_c \\ 0, & \omega_c < |\omega| \le \pi \end{cases}$$

$$h_{lp}[n] = F^{-1}[H_{lp}(j\omega)]$$

$$= \frac{\sin[\omega_c(n-M/2)]}{\pi(n-M/2)}$$

$$= \frac{\omega_c}{\pi} \frac{\sin[\omega_c(n-M/2)]}{\omega_c(n-M/2)}$$

$$= \frac{\omega_c}{\pi} \cdot \text{sinc}[\omega_c(n-M/2)]$$

symmetric: $h_{lp}[n] = h_{lp}[M-N]$

Kaiser window:

$$w[n] = \begin{cases} I_0\left[\beta\left(1-\left[(n-\alpha)/\alpha\right]^2\right)^{\frac{1}{2}}\right] \Big/ I_0(\beta), & 0 \le n \le M. \\ 0, \text{ otherwise.} \end{cases}$$

$$\alpha = \frac{M}{2}$$

$I_0[\cdot]$ = Zeroth order Modified Bessel Function of First kind.

$$I_0[x] = 1 + \frac{0.25x^2}{(1!)^2} + \frac{(0.25x^2)^2}{(2!)^2} + \frac{(0.25x^2)^3}{(3!)^2} + \ldots$$

$\Delta\omega = \omega_2 - \omega_1 = \omega_s - \omega_p$

$A = -20\log\delta, \quad \delta = \min(\delta_1, \delta_2)$

$$\beta = \begin{cases} 0.1102\,(A-8.7), & A > 50 \\ 0.5842\,(A-21)^{0.4} + 0.07886\,(A-21), & 21 \le A \le 50 \\ 0, & A \le 20. \end{cases}$$

$$M = \frac{A-8}{2.285\,\Delta\omega}$$

$$\omega_c = \frac{\omega_p + \omega_s}{2}$$

$$= \frac{\omega_1 + \omega_2}{2}$$

Stopband attenuation $(A_s) = -20\log(\delta)$, $\delta = \min(\delta_1, \delta_2$

Transition width : $\Delta\omega = \omega_2 - \omega_1 = \omega_s - \omega_p$

$\Delta\omega$ = Main lobe width.

$$\Delta\omega = \frac{8\pi}{M} \Rightarrow Ex$$

$$\omega_c = \frac{\omega_p + \omega_s}{2}$$

N = filter length

M = filter Order $= N-1$ //

Ham For Hamming.

$$\boxed{\Delta\omega = \omega_s - \omega_c}$$

//

Chapter 6:

Design of IIR Filter

(i) IIR Filter Design using Impulse Invariance.

Analog Filter: $H_a(s) = \sum_{k=1}^{N} \frac{C_k}{s - P_k}$, C_k = constants, P_k = poles.

inverse Laplace Transform

$$h_a(t) = \sum_{k=1}^{N} C_k \cdot e^{P_k \cdot t}$$

$$h[n] = \sum_{k=1}^{N} C_k \cdot e^{P_k n T} \quad , \quad T =$$

ZT

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} h[n] z^{-n}$$

$$= \sum_{k=1}^{N} \frac{C_k}{1 - e^{P_k T} \cdot z^{-1}}$$

Replace $s - P_k$ by $1 - e^{P_k T} z^{-1}$ //.

(2.) ~~Butt~~ Bilinear Transformation

⇒ Let : $H_a(s) = \frac{b}{s+a}$

$$H(z) = \frac{b}{\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+a}$$

$$\therefore \quad s = \frac{2}{T}\,\frac{1-z^{-1}}{1+z^{-1}}$$

$$= j\Omega$$

⇒ ~~$j\Omega = \frac{2}{T} j$~~

⇒ $\Omega = \frac{2}{T}\tan(\omega/2)$.

⇒ $\omega = 2\tan^{-1}\left(\frac{\Omega T}{2}\right)$.

Some commonly used Analog Filters & Their characteristics

(1) Butterworth (LPF) Filter

↳ simple, maximally-flat filter

* $|H(\Omega)|^2 = \dfrac{1}{1+\left(\dfrac{\Omega}{\Omega_c}\right)^{2N}}$

Magnitude Response.

$$= \frac{1}{1+\epsilon^2\left(\dfrac{\Omega}{\Omega_p}\right)^{2N}}$$

Ω_c = cut-off freq = -3dB freq.
N = Order of filter
ϵ = Scaling factor (≤ 1)

* Poles: $s_K = \Omega_c e^{j\pi/2} \cdot e^{j(2K+1)\pi/2N}$, $K = 0, 1, 2, \ldots, N-1$

* Order: $$N = \frac{\frac{1}{2}\log\left[\frac{1}{(1-\delta_p)^2} - 1\right] - \frac{1}{2}\log\left[\frac{1}{\delta_s^2} - 1\right]}{\log\left(\frac{\Omega_p}{\Omega_s}\right)}$$

$\left.\begin{aligned} \Omega_p &= \delta_p = \delta_1 \\ \Omega_s &= \delta_s = \delta_2 \end{aligned}\right\}$ not in dB.

OR!

$$N = \frac{\frac{1}{2}\log\left[\frac{10^{0.1A_s} - 1}{10^{0.1A_p} - 1}\right]}{\log\left(\Omega_s/\Omega_p\right)}$$

A_p = Passband ripple / attenuation.
A_s = Stopband "

$$\Omega_c = \frac{\Omega_p}{\left[\frac{1}{(1-\delta_p)^2} - 1\right]^{\frac{1}{2N}}}$$

$$H(s) = H_a(s) = \frac{\Omega_c^N}{(s-s_0)(s-s_1)\cdots}$$

Chebyshev (LPF) Filter:

→ 2 types → (i) Chebyshev = Chebyshev - Type I
(ii) Inverse " " " " II

Type - I:

* $|H(\Omega)|^2 = \frac{1}{1+\epsilon^2 T_N^2(\Omega/\Omega_p)}$

where,

. $T_N(x) = \begin{cases} \cos(N\cos^{-1}x), & |x| \le 1 \\ \cosh(N\cosh^{-1}x), & |x| > 1 \end{cases}$

. $\epsilon = \left[\frac{1}{(1-\delta_p)^2} - 1\right]^{\frac{1}{2}}$

Poles: $x_k + j y_k$, $k = 0$ to $(N-1)$

where,

$$x_k = r_2 \cos\phi_k$$
$$y_k = r_2 \sin\phi_k$$

where,

$$r_2 = \Omega_p \cdot \frac{\beta^2 - 1}{2\beta^2}, \quad \beta = \left(\frac{\sqrt{1+\epsilon^2} + 1}{\epsilon} \right)^{\frac{1}{N}}$$

$$\phi_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N} \text{ rad.}$$